SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA0716	**Course Title:**	COMPUTER NETWORKS
CO Assessed:	CO1	**Assessment:**	ASSESSMENT TOOL 1- SHORT ANSWER TEST
Weightage:	40%	**Total Marks:**	Out of 100
CO5 Statement:	***CO1:*** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*		
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Code of Conduct:

I, SUMETHA SRI.S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SUMETHA

ANNEXURE A

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3) Assessment Tool Weightage: 40%*

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

MARKS ALLOCATION

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (10%)	Total (100)
1							
2							
3							

4							
5							
6							
7							
8							
9							
10							
TOTAL							

1. File Transfer with Transport and Data Link Layers

Given:

- File size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**

Step 1: Convert file size to bytes

$150\ MB = 150 \times 10^6 = 150{,}000{,}000\ bytes$150 \text{ MB} = 150 \times 10^6 = 150{,}000{,}000 \text{ bytes}150 MB=150×106=150,000,000 bytes

Step 2: Calculate number of segments

$\text{Segments} = \frac{150{,}000{,}000}{1500} = 100{,}000$\text{Segments} = \frac{150{,}000{,}000}{1500} = 100{,}000Segments =1500150,000,000 =100,000

Step 3: Calculate number of frames

Each segment becomes one frame.

$\text{Frames} = 100{,}000$\text{Frames} = 100{,}000Frames=100,000

Step 4: Data Link Layer overhead

$\text{Overhead} = 100{,}000 \times 40 = 4{,}000{,}000\ bytes = 4{,}000{,}000\ \text{bytes}$\text{Overhead} = 100{,}000 \times 40 = 4{,}000{,}000 \text{ bytes} = 4{,}000{,}000 \text{ bytes}Overhead=100,000×40 =4,000,000 bytes=4,000,000 bytes=4,000,000 bytes

Step 5: Total transmitted data

$150{,}000{,}000 + 4{,}000{,}000 = 154{,}000{,}000\ bytes = 154{,}000{,}000\ \text{bytes}$150{,}000{,}000 + 4{,}000{,}000 = 154{,}000{,}000 \text{ bytes} = 154{,}000{,}000 \text{ bytes}150,000,000+4,000,000 =154,000,000 bytes=154,000,000 bytes=154,000,000 bytes

Convert to bits:

$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$154,000,000×8 =1,232,000,000 bits=1,232,000,000 |text{ bits}=1,232,000,000 bits

Step 6: Transmission Time

$\text{Time} = \frac{1,232,000,000}{50 \times 10^6}$Time=50×1061,232,000,000 =24.64 seconds=24.64 |text{ seconds}=24.64 seconds

Answer:

- Number of segments = **100,000**
- Number of frames = **100,000**
- Total overhead = **4,000,000 bytes (4 MB)**
- Transmission time = **24.64 seconds**

Explanation:

The **Transport Layer** divides data into segments, numbers them, and ensures reliable delivery using acknowledgments and retransmissions. The **Data Link Layer** encapsulates segments into frames, performs error detection using CRC, and controls node-to-node communication to provide reliability.

2. Sliding Window Protocol

Given:

- Window size = 6 frames
- Total frames = 48
- Frame size = 1024 bytes

Step 1: Calculate number of windows

$\text{Windows} = \frac{48}{6} = 8$Windows=648 =8

Step 2: Retransmissions

One frame is lost in every window.

$Retransmissions = 8 \times 1 = 8$ |$text{Retransmissions}=8$ |$times$ 1=8Retransmissions=8×1=8

Answer:

- Number of windows = **8**
- Total retransmissions = **8 frames**

Explanation:

Flow control prevents the sender from transmitting more data than the receiver can process. This reduces buffer overflow, improves efficiency, minimizes packet loss, and ensures smooth communication.

3. IP Packet Fragmentation

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

Step 1: Maximum data per fragment

$1500 - 20 = 1480$ bytes$1500 - 20 = 1480$ |$text{ bytes}$1500−20=1480 bytes

Step 2: Number of fragments

$\frac{12000}{1480} = 8.11$|$frac{12000}{1480}=8.11$148012000 =8.11

Ceiling value:

$= 9$ fragments$=9$ |$text{ fragments}$=9 fragments

Step 3: Header overhead

$9 \times 20 = 180$ bytes9 |$times 20 = 180$ |$text{ bytes}$9×20=180 bytes

Answer:

- Fragments created = **9**
- Total header overhead = **180 bytes**

Explanation:

The **Network Layer** divides oversized packets into fragments, assigns fragment offsets and identification fields, and reassembles them at the destination. If fragments are lost, higher-layer protocols request retransmission.

4. Sliding Window Protocol (Same as Question 2)

Given:

- Window size = 6
- Frames transmitted = 48

Number of windows

$48/6=848/6 = 8$48/6=8

Retransmissions

$8×1=88 \mid times\ 1=88×1=8$

Answer:

- Windows required = **8**
- Retransmissions = **8 frames**

Explanation:

Flow control synchronizes sender and receiver speeds, improving throughput and avoiding congestion.

5. Session Layer Synchronization

Given:

- Video file = 600 MB
- Checkpoint every = 60 MB
- Failure after = 390 MB

Step 1: Checkpoints before failure

$390/60=6.5390/60=6.5$390/60=6.5

Completed checkpoints:

$66$6

Checkpoint positions:

60, 120, 180, 240, 300, 360 MB

Step 2: Retransmitted data

Failure occurs at 390 MB.

Last checkpoint = 360 MB

$390−360=30MB390\text{-}360 = 30MB$390−360=30MB

Answer:

- Checkpoints created = **6**
- Data retransmitted after recovery = **30 MB**

Explanation:

Synchronization points allow data transmission to resume from the last checkpoint instead of restarting the entire session, saving time and bandwidth.

6. Presentation Layer Compression and Encryption

Given:

- Original file = 900 MB
- Compression = 40%
- Encryption overhead = 5%

Step 1: Compressed size

Remaining size:

$900 \times (1 - 0.40) 900 \, \backslash times \, (1\text{-}0.40)$900×(1−0.40) $= 900 \times 0.60 = 900 \, \backslash times \, 0.60$=900×0.60
$= 540 MB = 540 \, MB$=540MB

Step 2: Encrypted size

$540 \times 1.05540 \, \backslash times \, 1.05$540×1.05 $= 567 MB = 567 \, MB$=567MB

Step 3: Reduction percentage

$900 - 567900 \times 100 \backslash frac\{900\text{-}567\}\{900\} \backslash times 100$900900−567 ×100 $= 37\% = 37 \backslash \%$=37%

Answer:

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Explanation:

The Presentation Layer improves efficiency through compression and enhances security through encryption, ensuring confidentiality during transmission.

7. FTP Data Transfer

Given:

- Data = 3 GB
- Throughput = 120 Mbps
- Request size = 3 MB

Step 1: Convert data

$3GB = 3000MB$3GB=3000MB

Step 2: Number of requests

$30003 \backslash frac\{3000\}\{3\}$33000 $= 1000 = 1000$=1000

Step 3: Transmission time

Convert data to bits:

$3000 \times 83000 \backslash times\ 8$3000×8 $= 24000Mb = 24000\ Mb$=24000Mb
$Time = 24000120 \backslash text\{Time\} = \backslash frac\{24000\}\{120\}$Time=12024000 $= 200\ seconds = 200$
$\backslash text\{\ seconds\}$=200 seconds

Answer:

- Requests generated = **1000**
- Transmission time = **200 seconds**

Explanation:

The Application Layer provides services such as file transfer, authentication, error reporting, and user interaction through protocols like FTP.

8. End-to-End Delay

Given:

Layer	Delay
Application	4 ms
Transport	3 ms
Network	5 ms
Data Link	2 ms
Physical	1 ms

Propagation Delay = 18 ms

Transmission Delay = 12 ms

Step 1: Processing delay

*4+3+5+2+14+3+5+2+1*4+3+5+2+1 *=15ms=15ms*=15ms

Step 2: Total delay

*15+18+1215+18+12*15+18+12 *=45ms=45ms*=45ms

Answer:

- Total end-to-end delay = **45 ms**

Explanation:

Each OSI layer performs a specific function:

- Application: User services
- Transport: Reliable delivery
- Network: Routing
- Data Link: Framing and error detection
- Physical: Signal transmission

Together they contribute to overall delay.

9. Frame Efficiency Calculation

Given:

- Useful data = 80 MB
- Frame size = 1600 bytes
- Overhead = 80 bytes/frame

Step 1: Payload per frame

$1600 - 80 = 1520 \text{ bytes}$1600−80=1520 bytes

Step 2: Number of frames

$80{,}000{,}000 / 1520$80,000,000/1520 $= 52{,}631.58 = 52{,}631.58$=52,631.58 $= 52{,}632 \text{ frames} = 52{,}632 \text{ frames}$=52,632 frames

Step 3: Total overhead

$52{,}632 \times 80$52,632×80 $= 4{,}210{,}560 \text{ bytes} = 4{,}210{,}560 \text{ bytes}$=4,210,560 bytes

Step 4: Efficiency

$\frac{1520}{1600} \times 100$16001520 ×100 $= 95\% = 95\%$=95%

Answer:

- Frames required = **52,632**
- Total overhead = **4,210,560 bytes**
- Transmission efficiency = **95%**

Explanation:

Protocol overhead consumes bandwidth that could otherwise carry useful data. Excessive overhead reduces throughput and increases transmission costs.

10. Healthcare IoT Data Transfer

Given:

- Original data = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Frame overhead = 60 bytes

Step 1: Compressed file size

$240 \times (1 - 0.30)240|times(1-0.30)$240×(1−0.30) $= 168 MB = 168 \, MB$=168MB $= 168{,}000{,}000 \text{ bytes} = 168{,}000{,}000 \, \text{bytes}$=168,000,000 bytes

Step 2: Number of segments

$168{,}000{,}000 / 1400168,000,000/1400$168,000,000/1400 $= 120{,}000 = 120{,}000$=120,000

Step 3: Protocol overhead

$120{,}000 \times 60120,000 |times 60$120,000×60 $= 7{,}200{,}000 \text{ bytes} = 7{,}200{,}000 \, \text{bytes}$=7,200,000 bytes

Step 4: Total data transmitted

$168{,}000{,}000 + 7{,}200{,}000168,000,000 + 7,200,000$168,000,000+7,200,000 $= 175{,}200{,}000 \text{ bytes} = 175{,}200{,}000 \, \text{bytes}$=175,200,000 bytes

Answer:

- Compressed file size = **168 MB**
- Number of segments = **120,000**
- Protocol overhead = **7,200,000 bytes (7.2 MB)**
- Final data transmitted = **175.2 MB**

Explanation:

Multiple OSI layers collaborate to ensure secure and reliable communication:

- **Application Layer:** Provides healthcare service access.

- **Presentation Layer:** Compresses and encrypts patient records.
- **Session Layer:** Maintains communication sessions.
- **Transport Layer:** Segments data and ensures reliable delivery.
- **Network Layer:** Determines routing paths.
- **Data Link Layer:** Frames data and performs error detection.
- **Physical Layer:** Transmits bits through the communication medium.

This layered approach guarantees confidentiality, integrity, reliability, and efficient transmission of sensitive medical information.